Percent Change Bar Model

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Percent Change Bar Model** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A $50 jacket is marked up 20%. Then the new price is discounted 20%. Is the final price $50?
   No: 50 × 1.2 = $60, then 60 × 0.8 = $48.
2. What single number do you multiply by to increase a price by 25%?
   1.25

2 Explore use the tool

1. Original $80, increase 25%. What is each 10% section worth, and what is the new amount?
   Each section is $8. The new amount is $100 (125% of $80).
2. Increase $80 by 20%, then decrease by 20%. Where do you end up, and why?
   $76.80. The 20% decrease is 20% of $96 ($19.20), which is more than the $16 added.
3. After a 25% increase, which second change brings the price exactly back to the start?
   A 20% decrease ($100 × 0.8 = $80).
4. Find two increases in a row that make a total increase of exactly 32%.
   +10% then +20% (1.1 × 1.2 = 1.32), or +20% then +10%.

3 Record use the tool and fill in the table

Percent changes

Use the second change when there are two.

Original	Change(s)	Multiply by	Final amount	Overall change
$60	−15%	0.85	$51.00	−15%
$40	+30%, then −10%	1.3 × 0.9 = 1.17	$46.80	+17%
$120	−25%, then +25%	0.75 × 1.25 = 0.9375	$112.50	−6.25%
$200	+10%, then +10%	1.1 × 1.1 = 1.21	$242.00	+21%

4 Explain what did you discover?

1. A p% increase multiplies by $1 + \frac{p}{100}$. A p% decrease multiplies by $1 - \frac{p}{100}$.
2. In a second change, the percent is a percent of the **new** amount.
3. Two percent changes in a row **multiply**; they do not add.
4. Explain why an increase of 50% followed by a decrease of 50% does not get you back to the original amount.
   The 50% decrease is half of the bigger amount: 100 → 150 → 75. 1.5 × 0.5 = 0.75, an overall 25% decrease.

5 Apply a new situation, no tool

1. A $36 meal has a 15% tip added. What is the total?
   $41.40
2. A $250 bike is 30% off. Then 6% tax is added to the sale price. What is the final cost?
   $250 × 0.7 × 1.06 = $185.50
3. A price went from $40 to $46. What is the percent change?
   +15%
4. After a 20% discount a shirt costs $32. What was the original price?
   $40 ($32 ÷ 0.8)

Teacher notes

Intended discovery: A percent change multiplies by (100% ± p%) of the starting amount; in a chain the base changes, so changes multiply rather than add.

Likely misconceptions:

- +p% then −p% returns to the start.
- Adding percents in a row (+10% and +20% = +30%).
- Taking the discount percent of the sale price instead of the original (Apply 4).

Answers that can vary: Try this 2 (many original/percent pairs add $24) and Explore 4 (order of the two increases).